Experimental probability

1 A retail store served 773 customers in October, and there were 44 complaints during that month.
Calculate the experimental probability that a customer complains. Give your answer as a percentage, rounded to the nearest whole percent.

2 At a factory, 1000 computers were tested with 15 found to be faulty. Calculate the experimental probability that a computer at this factory will be faulty.

3 Find the experimental probability of the following:

a Flipping heads from a coin that was flipped 184 times with 93 heads recorded.

b A computer being faulty at a factory where 1000 computers were tested and 15 were found to be faulty.

c A car driving by is white, given that 1919 cars had gone by and 77 of them were white.

4 At a set of traffic lights, the green light is on for 24 seconds, then the yellow light lasts for 3 seconds, and then the red light is on for 13 seconds. This cycle then repeats continuously. If a car approaches the traffic light, calculate the probability that the light will be:

a Green b Yellow c Red

5 An insurance company found that in the past year, of the 2558 claims made, and 1493 of them were from drivers under the age of 25. Calculate the experimental probability, as a whole percentage, that a claim is filed by someone:

a Under the age of 25. b 25 years or older.

6 The experimental probability that a person uses public transport is 50%. Out of 500 people, how many would you expect to use public transport?

7 Hermione rolled a standard six-sided die 60 times.

a State the theoretical probability of rolling a six.

b How many times would she expect a six to appear?

c After she finished rolling the die, she noticed that she had rolled a six 24 times. Find the experimental probability of getting a six.

d Was the theoretical and experimental probability of getting a six the same? Explain your answer.

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8 A factory produces tablet computers. In March it produced 8000 tablets, and 240 were found to be faulty.

a Calculate the experimental probability that a tablet produced by the factory is faulty.

b The factory plans to produce 9000 tablets in April. How many should they expect to be faulty?

9 Uther flipped a coin 14 times.

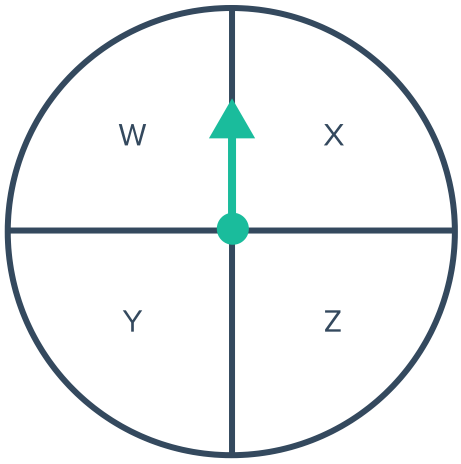
a How many times would he expect a tails to appear?

b After flipping the coins, he noticed that tails had appeared 4 times. Find the experimental probability of getting a tails.

10 Derek spun the following spinner 20 times:

a How many times would he expect the arrow to land on X?

b After he finished spinning, he noticed that the arrow fell on X 8 times. Find the experimental probability of getting an X.



11 Oprah has a bag with 2 red balls, 2 blue balls, and 2 green balls in it. She took a ball out of the bag and returned it 24 times.

a How many times would she expect to get a green ball?

b After she finished, she noticed that she had drew a green ball 18 times. Find the experimental probability of getting a green ball.

12 Georgia is drawing a card out of a deck of 10 cards, labeled from 1 to 10. She drew a card and returned it to the bag 40 times.

a How many times would she expect to get the card with 6 on it?

b After she finished, she noticed that the card with 6 on it was drawn out of the bag 21 times. Find the experimental probability of getting a card with 6 on it.

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13 Of 50 people buying coffee at a cafe, 39 bought a latte.

a Calculate the fraction of people that bought a latte.

b In one day, the cafe had 550 customers. How many would you expect to order a latte?

14 A maternity ward delivers the following number of babies each day for two weeks:

$$5, 6, 4, 7, 6, 6, 4, 6, 5, 5, 6, 5, 6, 6$$

a How many babies can you expect the ward to deliver on any particular day?

b How many babies would you expect the ward to deliver in a year?

15 Of 3000 students graduating as teachers from a particular university, 8% graduate as Mathematics teachers.

a How many students graduated as Mathematics teachers from this university?

b In the country, there are 6900 new graduate teachers. How many are likely to be Mathematics teachers?

Tables and graphs

16 Boxes of individually wrapped chocolates, with 20 chocolates per box, are randomly sampled to see how many blueberry chocolates are in each box.
The results from the samples are shown in the table:

Box No.	1	2	3	4	5	6	7	8	9	10
Blueberry bites	0	1	2	1	2	3	3	0	3	4

a How many samples were taken?

b Calculate the fraction of blueberry chocolates in box 2.

c How many blue berry chocolates would you expect to be in each box?

17 To prepare for the week ahead, a restaurant keeps a record of the number of each main meal ordered throughout the previous week.

a How many meals were ordered altogether?

b Calculate the experimental probability that a customer will order a beef meal. Give your answer as a percentage, rounded to the nearest whole percent.

Meal	Frequency
Chicken	54
Beef	32
Lamb	26
Vegetarian	45

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18 The following table shows the outcomes of tossing three coins 102 times:
Find the experimental probability of:

Outcome	Frequency
HHH	11
HHT	12
HTH	11
HTT	16
THH	12
THT	15
TTH	10
TTT	15

a Tossing 3 tails.

b Tossing at least 2 heads.

c Tossing at least 1 tail.

d Tossing only 1 head.

e Tossing exactly 2 tails.

19 Boxes of toothpicks are examined and the number of toothpicks in each box is recorded in the table shown:
If the number of toothpicks of another box were counted, find the experimental probability it will:

Number of toothpicks	Number of Boxes
87	0
88	6
89	4
90	1
91	1
92	2
93	1

a Contain 89 toothpicks.

b Contain more than 90 toothpicks.

c Contain less than 90 toothpicks.

20 High school students attending an international conference were asked to register what language they speak other than English. The results are shown in the table below:

Language	Frequency
French	20
Arabic	13
Spanish	21
Mandarin	19
Other	37

a How many students attended the conference?

b Find the probability that a student chosen at random speaks:

i French

ii Mandarin

iii Arabic or Spanish.

iv Spanish or Other.

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21 A die is rolled 60 times and the results are recorded in the table:

Number	Frequency
1	8
2	10
3	8
4	10
5	10
6	14

a Calculate the relative frequency of rolling a 6.

b Calculate the relative frequency of rolling a 3 or higher.

c Calculate the relative frequency of rolling a 3 or lower.

22 The table shows the results of rolling a die:

Outcome	Frequency
1	13
2	12
3	20
4	14
5	15
6	16

a How many times was the die rolled?

b What is the experimental probability of rolling a 2?

c What is the experimental probability of rolling an even number?

23 Five schools compete in a basketball competition. The results from the last season are given in the table below:

Schools competing each game	Winner
St Trinian's vs Ackley Bridge	St Trinian's
Ackley Bridge vs Summer Heights	Summer Heights
Lakehurst vs Marquess	Marquess
Marquess vs St Trinian's	St Trinian's
St Trinian's vs Summer Heights	Summer Heights
St Trinian's vs Lakehurst	Lakehurst
Ackley Bridge vs Marquess	Ackley Bridge
Marquess vs Summer Heights	Summer Heights
Lakehurst vs Summer Heights	Lakehurst
Ackley Bridge vs Lakehurst	Lakehurst

Calculate the experimental probability that Ackley Bridge wins one of their matches.

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24 The table shows the main fighting style of each fighter in a mixed martial arts tournament:

Style	Frequency
Karate	40
Wrestling	54
Judo	47
Taekwondo	59

a Calculate the total number of fighters.

b Calculate the relative frequency of a fighter using Karate.

c Calculate the experimental probability of a fighter using wrestling.

d Calculate the probablity that a fighter does Judo.

e Calculate the experimental probability of a Taekwondo fighter competing.

f Which is the most common fighting style?

g If 100 further tournaments were to be held, which fighting style would you expect to be the least used?

25 The table shows the number of trains that arrived on time at the local station during the week:

Day	Number of trains	On time
Monday	29	22
Tuesday	24	23
Wednesday	21	19
Thursday	27	22
Friday	23	20

a Calculate the experimental probability that a train will be on time on Monday, as a whole percentage.

b Which day had the highest experimental probability of a train being on time?

c Calculate the experimental probability of a train arriving on time across the entire week, as a whole percentage.

26 In an experiment, several balls coloured either green or black and numbered 1 to 11 are placed in a bag. The table shows all the outcomes that occured:

	1	2	3	4	5	6	7	8	9	10	11
Green	×	×	×	×	×	×	×	×	×	×	×
Black	×	×	×	×	×	×					

Find the probability that a ball drawn at random:

a Is black.

b Has the number 2.

c Has the number 7.

d Has a number 3 or higher.

e Is green and has the number 5.

f Is green or has the number 5.

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27 At a golf course, the number of golf balls lost each day over several days were recorded, and rounded to the nearest 10. The table shows the results:
Using the data, estimate the probability that more than 39 golf balls will be lost tomorrow.

Golf Balls Lost	Days
10	5
20	5
30	1
40	4
50	3

28 A group of students were surveyed on their eye colour. The results are shown in the table:

a Calculate the percentage of students that had hazel eyes.

b Calculate the least number of students that could have completed the survey.

Brown	Blue	Green	Hazel
70%	15%	2%	

29 The following table shows the number of people who scored a certain mark on a recent test. The test is marked out of 50.

a Calculate the probability that someone scored more than 50% on the test.

b Calculate the probability that someone scored between 21 and 40 inclusive.

Marks %	Frequency
$11 - 15$	5
$16 - 20$	1
$21 - 25$	6
$26 - 30$	2
$31 - 35$	4
$36 - 40$	9
$41 - 45$	10
$46 - 50$	3
Total	40

30 The following table shows the weekly wage of a number of workers that were surveyed:

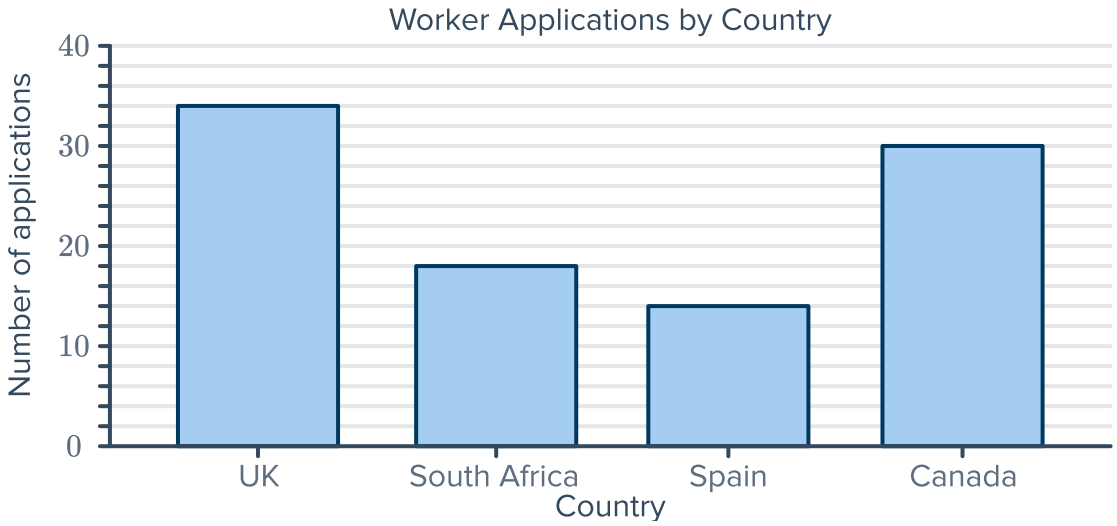
a Calculate the probability that a worker chosen from the sample has a wage of at least $\$280$ per week.

b Calculate the probability that a worker chosen from the sample has a wage of more than $\$294$.

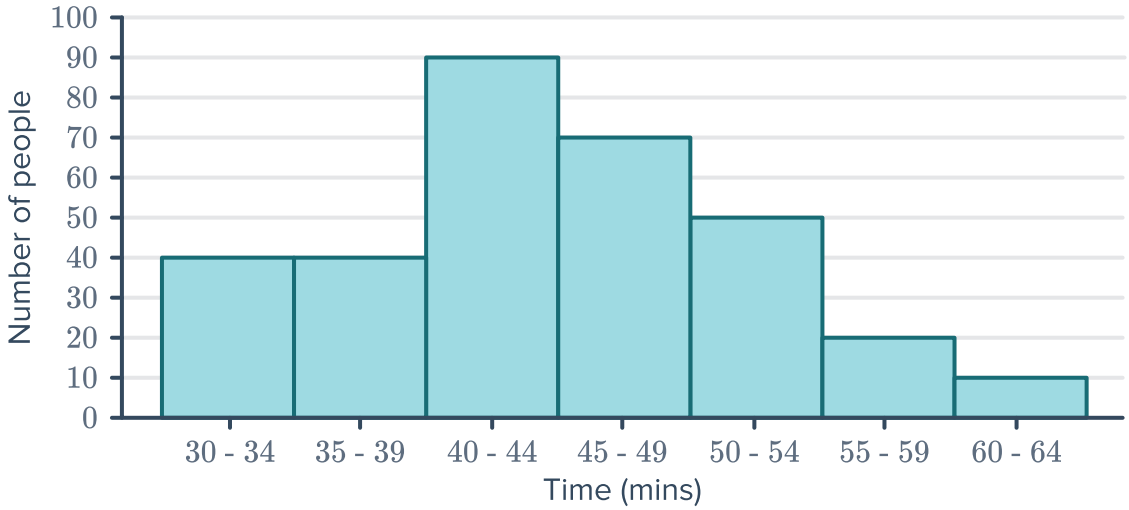
Weekly wages in dollars	Frequency
$220 - 234$	5
$235 - 249$	3
$250 - 264$	2
$265 - 279$	2
$280 - 294$	10
$295 - 309$	2
$310 - 324$	2
Total	26

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31 The graph shows the four countries that university students applied to for exchange in the last month. Calculate the relative frequency of the country with the fewest applications.



32 The following graph shows the number of people that were served at a furniture store, and the length of time it took to serve them:

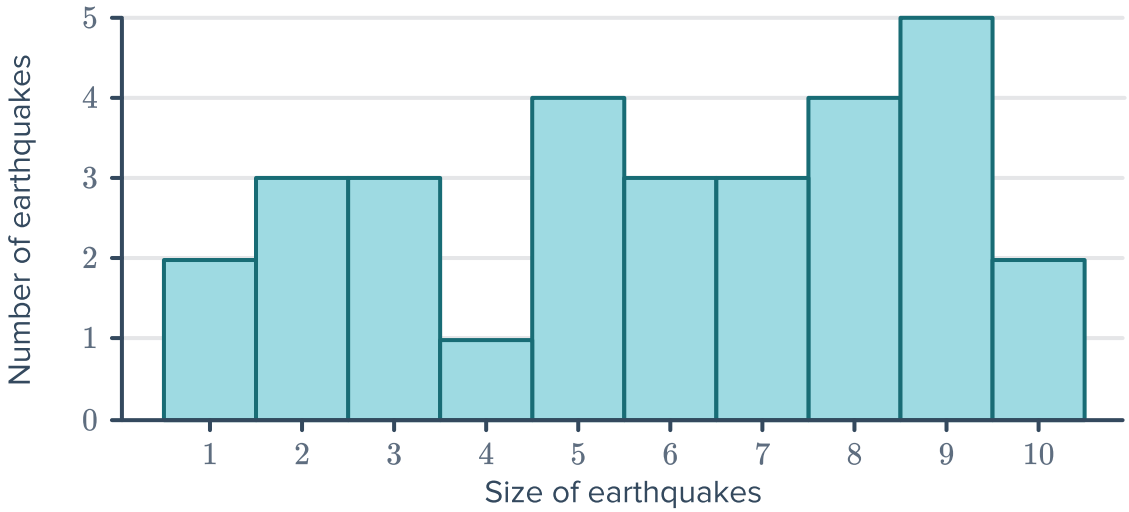


a Calculate the probability that someone was served in under 40 minutes.

b Calculate the probability that someone had to wait at least 50 minutes to be served.

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33 The size of several earthquakes was measured over a period of time and the results are presented in the graph. The numbers on the horizontal axis represent the class centres:



a Calculate the probability that an earthquake will measure $5, 6$, or 7.

b Calculate the probability that an earthquake measures less than 5.

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